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Department of Information Technology

A.Y. 2025-2026
Class: BE-IT A/B, Semester: VIII
Subject: Cloud Computing Lab

Student Name: Vishal R Mahajan **Student Roll No: 1**

Experiment – 2: Bare-metal Virtualization

Aim: To study and Implement Bare-metal Virtualization using Xen (Open Source tool), HyperV or VMware Esxi.

Objective: After performing the experiment, the students will be able to learn –
- Understand functionalities of OS and Hypervisor
- Working of Hypervisor
- Getting familiar with key concepts of virtualization
- Types of virtualization
- Usefulness of virtualization

Lab objective mapped : ITDO8024.1: To get familiar and implement different types of virtualization techniques.

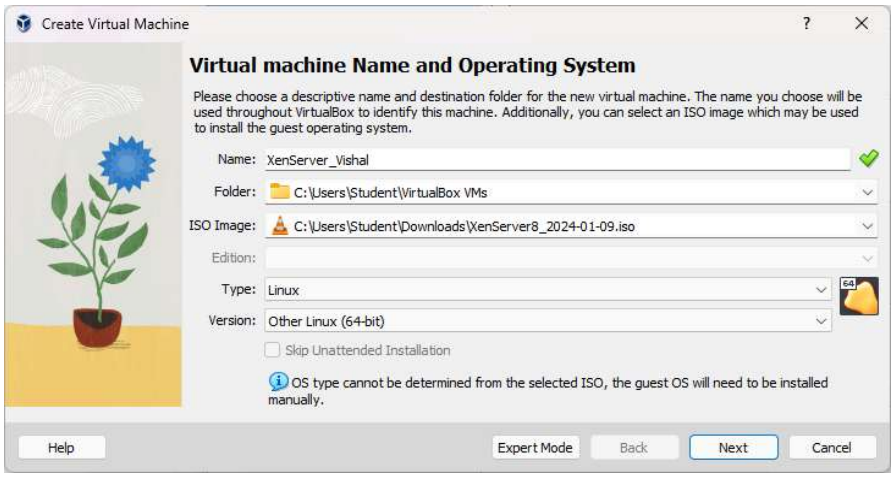
Prerequisite: An AWS account

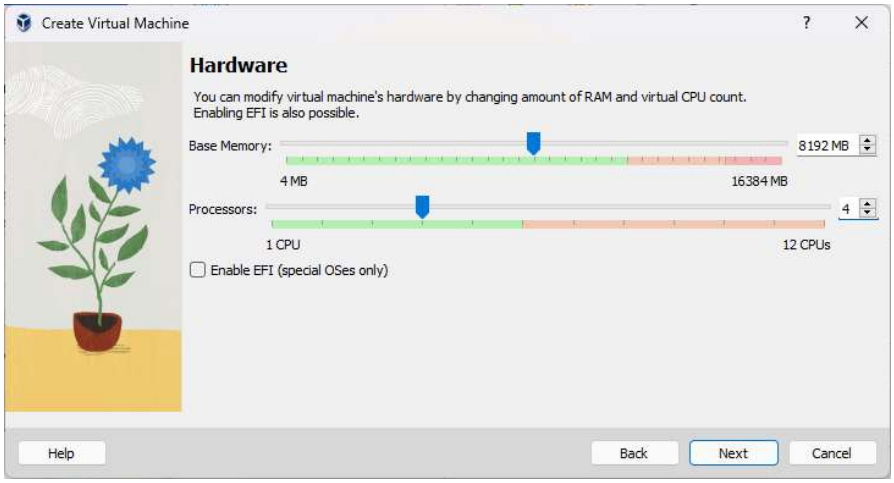
Requirements:
- XenServer ISO

Pre-Experiment Theory:
1. **Hypervisor** (description-To be written in hand))
2. **Bare-metal hypervisor Architecture-**(To be drawn in hand)
3. **Xen Architecture-**(To be drawn in hand)

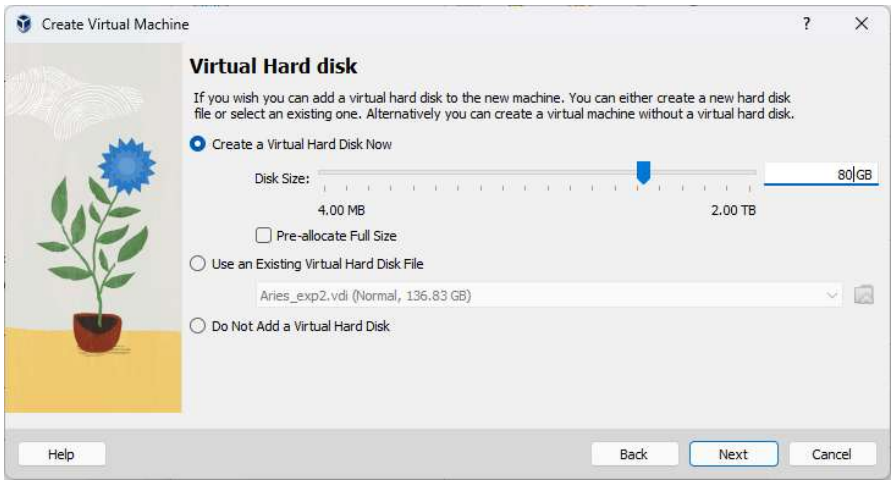
Procedure (Soft Copy form)
- **Install Xen Server**
  1. Download the Citrix Hypervisor (XenServer) .iso file Download
  2. Create an account to complete the download
  3. Launch Virtualbox
  4. Create a New VM by selecting Machine > New
     i. Name: XenServer
     ii. Machine Folder: C:\VMs
     iii. Type: Linux

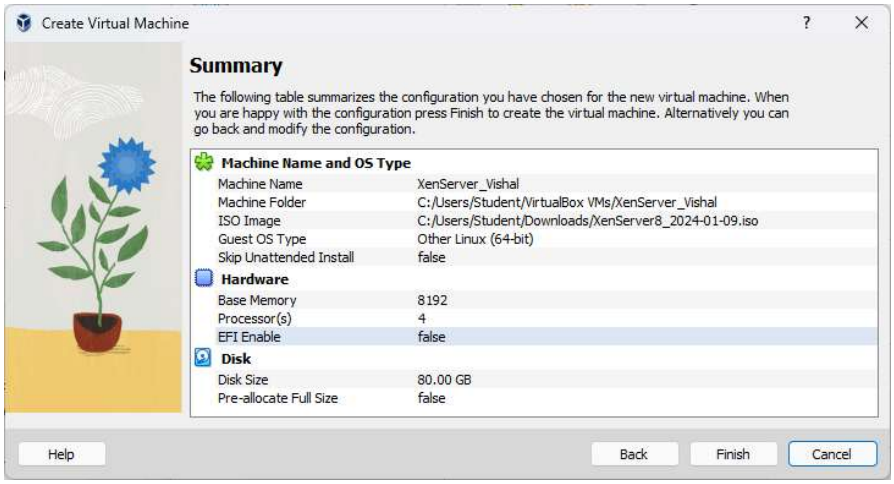
iv. Version: Other (64-bit)
v. Memory Size: 8192 MB
vi. Hard disk: Create a virtual hard disk now



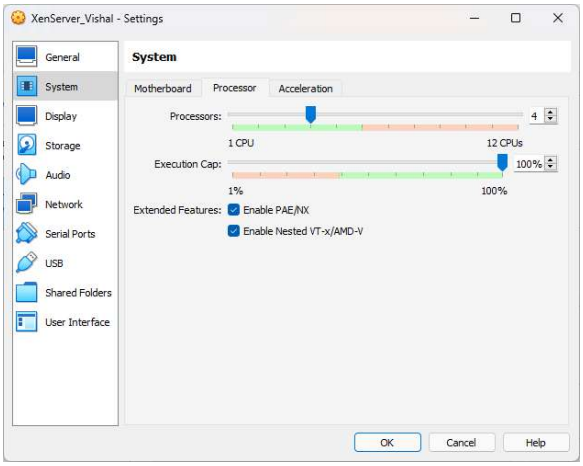


5. Click Create
6. On the Create Virtual Hard Disk dialog
   i. Name the virtual disk image XenServer.vdi
   ii. File size: 500 GB
   iii. Hard disk file type: VDI
   iv. Storage on physical hard disk: Dynamically Allocated

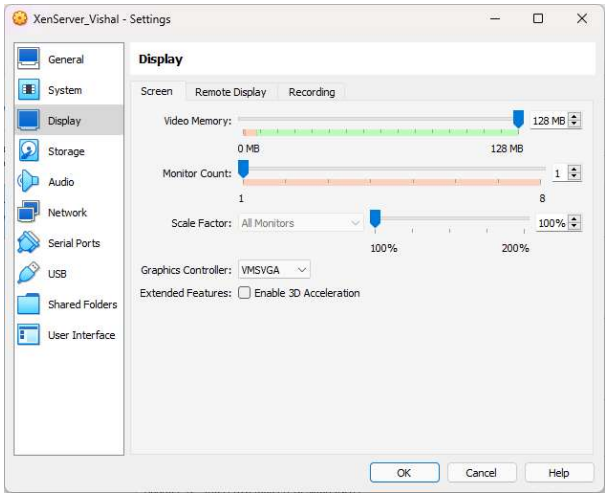




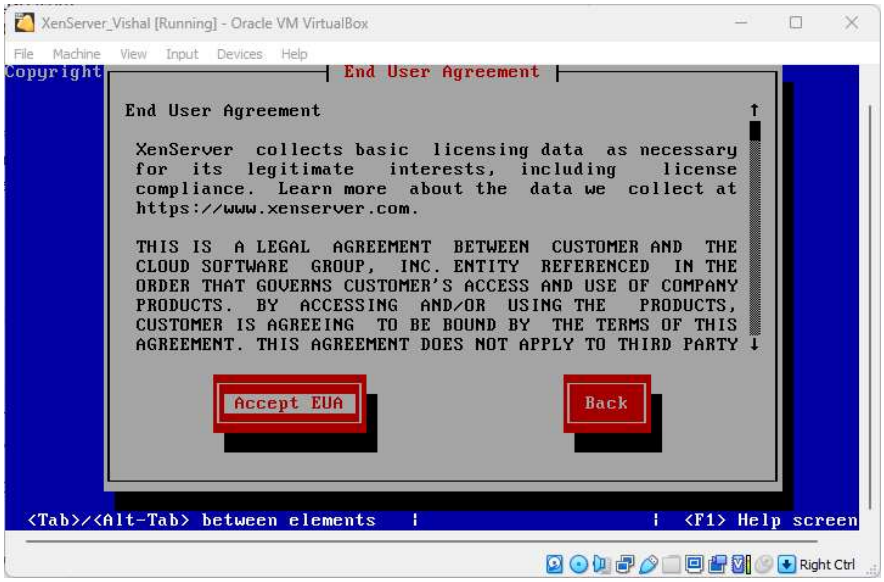
7. Click Create
8. Select the VM and Click Settings
9. Select System > Processor
10. Give the VM at least 2 processors
11. Check the Enable Nested VT-x/AMD-V



12. Select Display
13. Slide the Video Memory to 128 MB

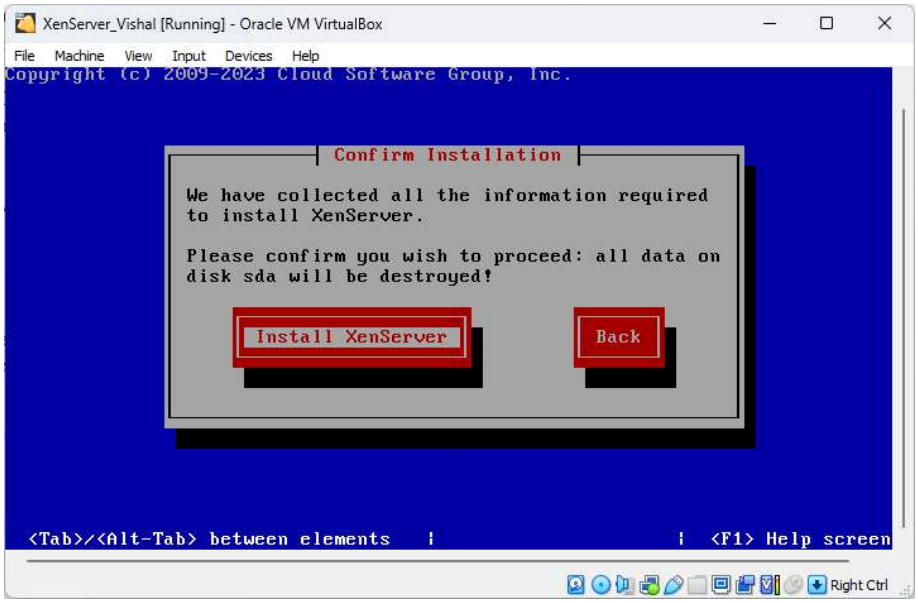


14. Select Network
15. Set the attached to dropdown to Bridged Adapter
16. Select Storage
17. Click on the CD-ROM drive
18. Select the disc dropdown to the right > Choose a virtual optical disc file...
19. Browse to and select the downloaded Citrix Hypervisor (XenServer) .iso file
20. Click OK
21. Select the hard disk dropdown to the right
22. Click the Create button at the top
23. Change the size to 500 GB > Click Create
24. Click OK to accept the settings
25. Make sure the XenServer VM is selected and click Start > Normal
26. Select a keyboard mapping > Ok
27. Select Ok to begin the installation
28. Select Accept EULA

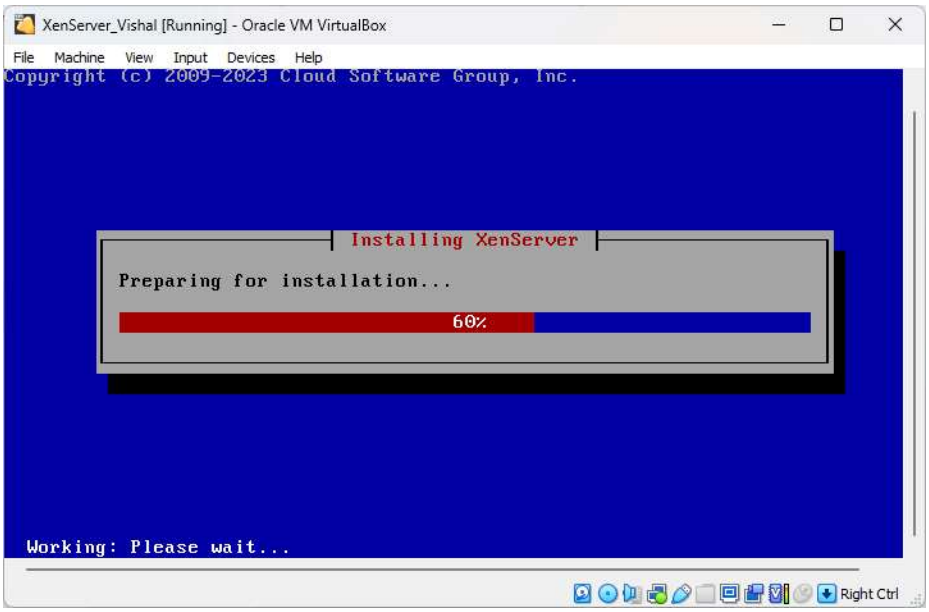


29. Press Spacebar to select the 500GB VBOX HARDDISK
30. Arrow down and check the Enable thin provisioning box

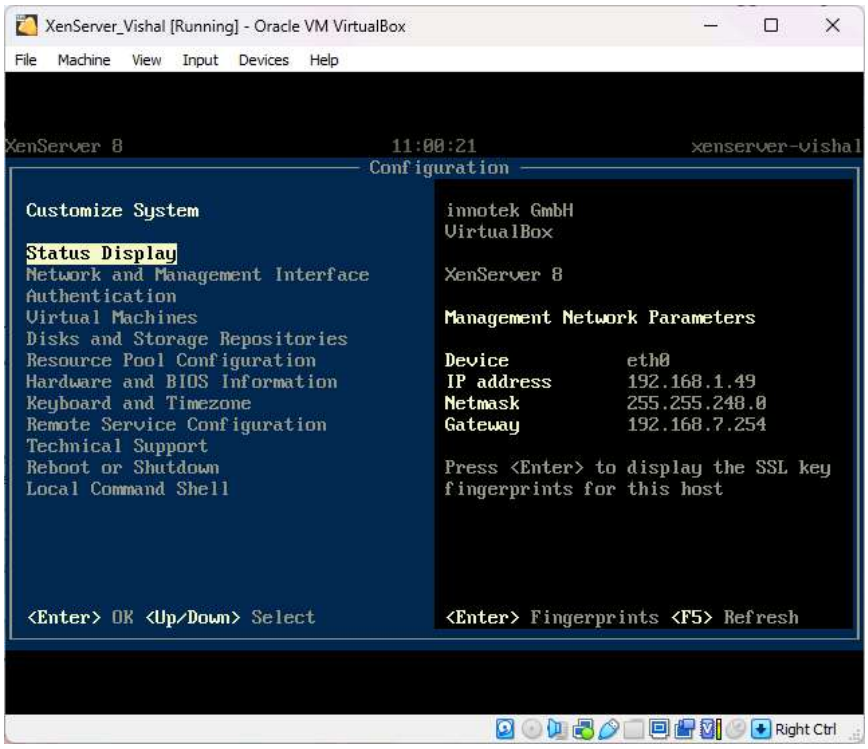
31. Select Ok
32. Select Local Media > Ok
33. Enter and confirm a root password
34. Configure the network > Ok
35. Specify a hostname > Ok
36. Set the time zone > Ok
37. Select Using NTP > Ok
38. Select Ok at the NTP server setup
39. Select Install Citrix Hypervisor



40. Select No at the supplemental packs installation screen



41. After the installation completes, select Devices > Optical Drives > Remove disk from virtual drive
42. Select Ok to reboot
43. Welcome to Citrix Hypervisor (XenServer) inside VirtualBox



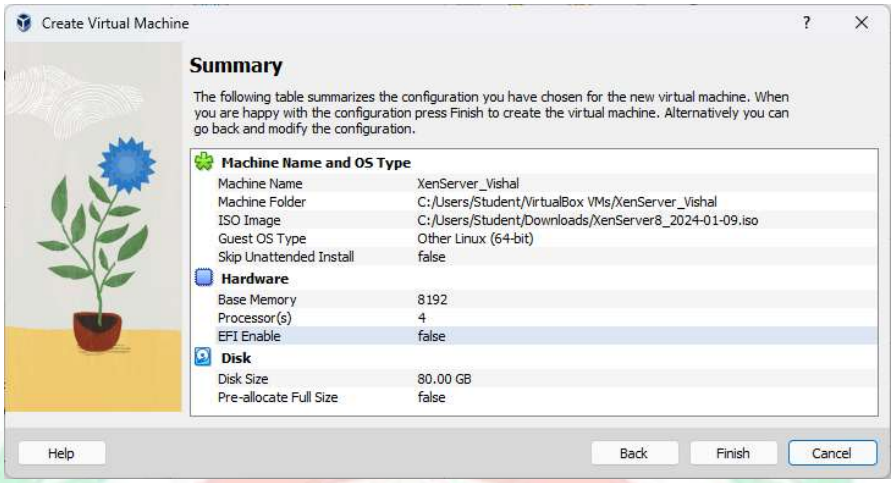
- **Installing Citrix XenCenter**
  1. Go back to the download page and download XenCenter Download
  2. Install the downloaded .msi
  3. Launch Citrix XenCenter
  4. Click the Add a Server option in the main window
  5. Enter the IP address and root password to the Citrix Hypervisor VM

- **To connect XenCenter to the Citrix Hypervisor server:**
  a. Launch XenCenter. The program opens to the Home tab.
  b. Click the Add New Server icon.
  c. Enter the IP address of the Citrix Hypervisor server in the Server field. Type the root user name and password that you set during Citrix Hypervisor installation. Click Add.
  d. The first time you add a host, the Save and Restore Connection State dialog box appears.
  e. This dialog enables to set preferences for storing host connection information and automatically restoring host connections.

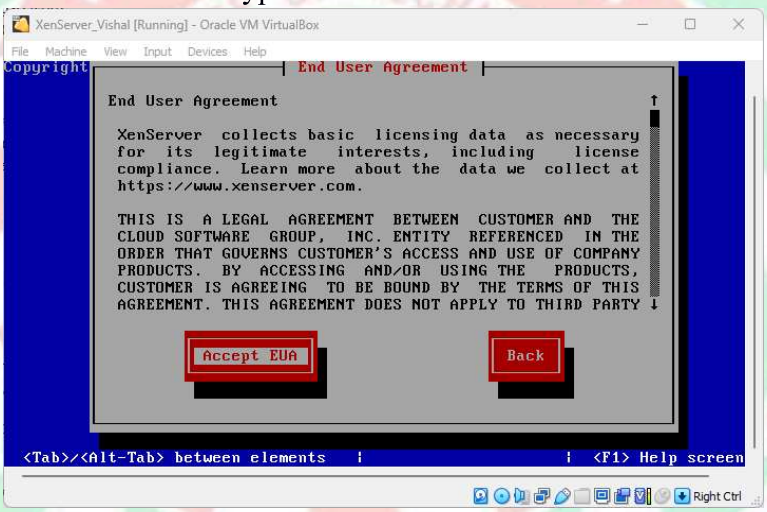
Post-Experiments Exercise

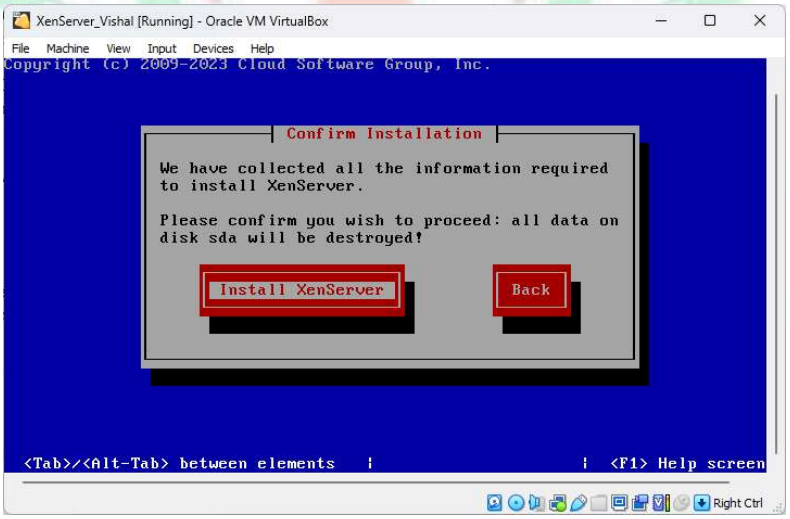
Attach Followin Screen Shots-

1. Creation of VM inside Oracle Box

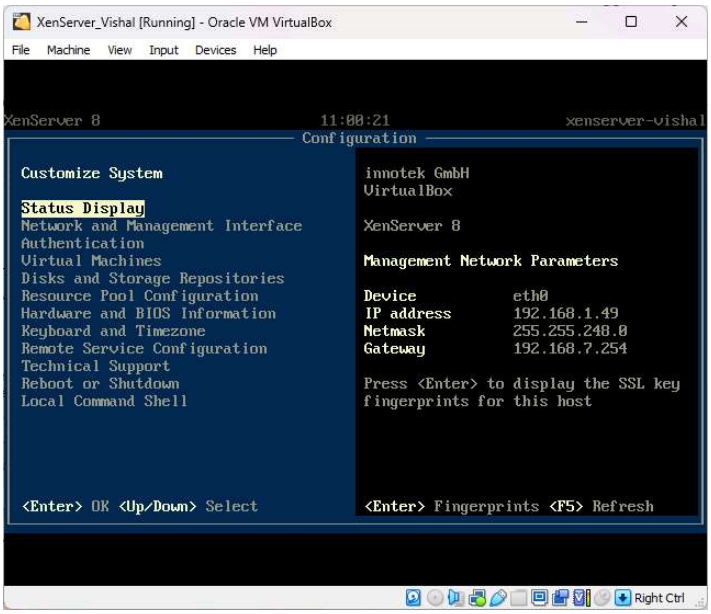


2. Any two screen shots of Xen hypervisor installation

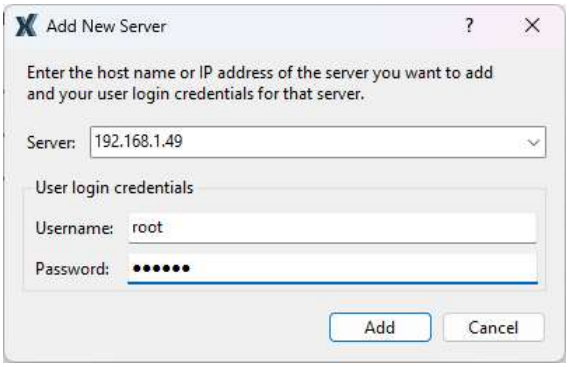


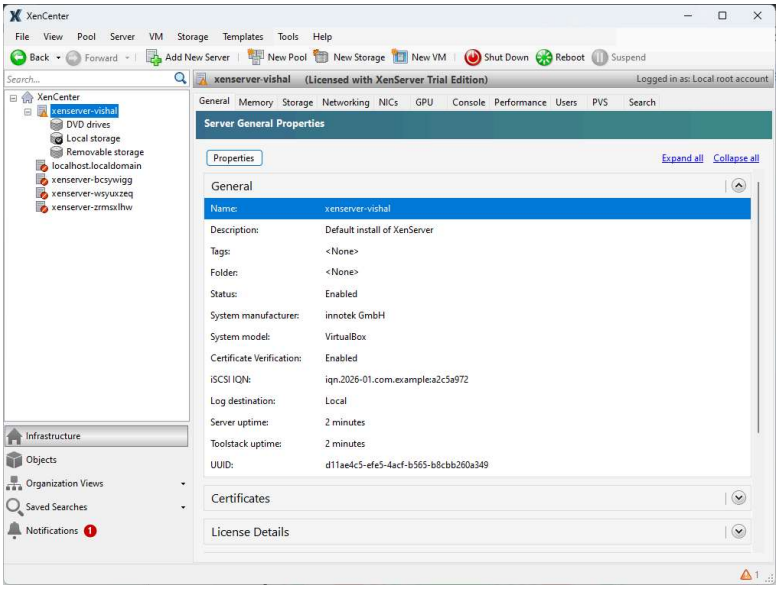


3. final working screen of xen hypervisor



4. Enter the IP address and root password to the Citrix Hypervisor VM from xen center





Extended Theory:

1. Difference between bare-metal and hosted hypervisors?(to be written in hand)
2. Bare-metal hypervisor use cases(to be written in hand)

3. Benefits and drawbacks of bare-metal hypervisors(soft copy)

Answer:

Benefits:

Bare-metal hypervisors provide very high performance because there is no host operating system layer consuming resources. They offer strong security since fewer software layers reduce attack surface. Stability and reliability are higher because the hypervisor is purpose-built for virtualization workloads. They support better resource management, allowing CPU, memory, and storage to be allocated efficiently among virtual machines. They are ideal for enterprise environments and data centers where uptime and scalability are critical.

Drawbacks:

They require specialized knowledge to install and manage, making setup more complex for beginners. Hardware compatibility can be limited compared to hosted hypervisors. Initial cost may be higher due to licensing and enterprise-grade hardware requirements. Troubleshooting can be harder because there is no full general-purpose operating system available for diagnostics. They are less convenient for casual users or small personal systems

4. Top bare-metal hypervisor vendors and products(soft copy)

VMware : VMware ESXi
Microsoft: Hyper V Server or Windows Server Hyper V
Citrix : Citrix Hypervisor formerly XenServer
Oracle : Oracle VM Server for x86
Red Hat : Red Hat Virtualization Hypervisor based on KVM
Nutanix : Nutanix AHV Acropolis Hypervisor

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

[1]Linux foundation: ”Xen tools”. Information available at:
http://wiki.xen.org/wiki/Xen_tools
Last review: 16-03-2014: 13.51
[2] Bill HILL “Intro to Virtualization: Hardware, Software, Memory, Storage, Data and Network Virtualization Defined”. Information available at: http://www.petri.co.il/intro-tovirtualization.htm
Last review: 16-03-2014: 12.55
[3] [Online]https://docs.citrix.com/en-us/citrix-hypervisor/install.html
[4] [Online]https://i12bretro.github.io/tutorials/0372.html